# Sequences & Series

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$**

Find the coefficient of $x^6$ in the binomial expansion of $(1+x)^{-3}$.

Coefficient $= -36 = 201606! = 28.$

**2** **[Foundational] • Target: Approximating Roots via Binomial Series**

Find the coefficient of $x^3$ in the binomial expansion of $(1+x)^{-1}$.

Coefficient $= -13 = -63! = -1.$

**3** **[Foundational] • Target: Method of Differences & Telescoping Series**

Find the coefficient of $x^5$ in the binomial expansion of $(1+x)^5$.

Coefficient $= 55 = 1205! = 1.$

**4** **[Foundational] • Target: Maclaurin Series & Higher-Order Approximations**

Find the coefficient of $x^2$ in the binomial expansion of $(1+x)^{-3}$.

Coefficient $= -32 = 122! = 6.$

**5** **[Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$**

Find the coefficient of $x^2$ in the binomial expansion of $(1+x)^4$.

Coefficient $= 42 = 122! = 6.$

**6** **[Foundational] • Target: Approximating Roots via Binomial Series**

Find the coefficient of $x^4$ in the binomial expansion of $(1+x)^2$.

Coefficient $= 24 = 04! = 0.$

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**7** [Foundational] • Target: Method of Differences & Telescoping Series

The rational function $f$ is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where $k$ is a real constant. Given that the series expansion of $f(x)$ in ascending powers of $x$, valid for sufficiently small $|x|$, has no term in $x^2$, find the value of $k$.

Final Answer: $\frac{25}{4}$. To find the series expansion of $f(x)$, we first decompose $f(x)$ into partial fractions. **Step 1: Algebraic Division** The degree of the numerator is $4$ and the degree of the denominator is $1 + 2 = 3$. Because the numerator has a higher degree than the denominator, $f(x)$ is an improper algebraic fraction and polynomial division must be carried out first. Expanding the denominator: $(x-2)(x^2+1) = x^3 - 2x^2 + x - 2$. Dividing $2x^4 - 3x^3 + kx^2 - 8x + 3$ by.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of $x^5$ in the Maclaurin series of $e^x$.

Final Answer: $\frac{1}{120}$. The Maclaurin series of $e^x$ gives coefficient $\frac{1}{5!} = \frac{1}{120}$ for $x^5$.

**9** [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

Find the coefficient of $x^6$ in the Maclaurin series of $e^x$.

Final Answer: $\frac{1}{720}$. The Maclaurin series of $e^x$ gives coefficient $\frac{1}{6!} = \frac{1}{720}$ for $x^6$.

**10** [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of $x^4$ in the Maclaurin series of $e^x$.

Final Answer: $\frac{1}{24}$. The Maclaurin series of $e^x$ gives coefficient $\frac{1}{4!} = \frac{1}{24}$ for $x^4$.

**11** [Intermediate] • Target: Method of Differences & Telescoping Series

Find the coefficient of $x^3$ in the Maclaurin series of $e^x$.

Final Answer: $\frac{1}{6}$. The Maclaurin series of $e^x$ gives coefficient $\frac{1}{3!} = \frac{1}{6}$ for $x^3$.

**12** [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of $x^5$ in the Maclaurin series of $\sin x$.

Final Answer: $\frac{1}{120}$. $\sin x = \sum (-1)^m \frac{x^{2m+1}}{(2m+1)!}$. Coefficient of $x^5$: $\frac{1}{120}$.

**13** [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

Find the coefficient of $x^6$ in the Maclaurin series of $\cos x$.

Final Answer: $-\frac{1}{720}$. $\cos x = \sum (-1)^m \frac{x^{2m}}{(2m)!}$. Coefficient of $x^6$: $-\frac{1}{720}$.

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**14** [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of $x^4$ in the Maclaurin series of $\cos x$.

Final Answer: $\frac{1}{24}$. $\cos x = \sum (-1)^m \frac{x^{2m}}{(2m)!}$. Coefficient of $x^4$: $\frac{1}{24}$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 7.01 Vectors

Find the coefficient of $x^4$ in the Maclaurin series of $\ln(1+x)$.

Final Answer: $-\frac{1}{4}$. $\ln(1+x) = \sum (-1)^{n+1} \frac{x^n}{n}$. Coefficient of $x^4$: $-\frac{1}{4}$.

**16** [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find the coefficient of $x^2$ in the Maclaurin series of $\ln(1+x)$.

Final Answer: $-\frac{1}{2}$. $\ln(1+x) = \sum (-1)^{n+1} \frac{x^n}{n}$. Coefficient of $x^2$: $-\frac{1}{2}$.

**17** [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$ • Cross-Lesson: 9.01 Complex Numbers

Find the coefficient of $x^6$ in the Maclaurin series of $\ln(1+x)$.

Final Answer: $-\frac{1}{6}$. $\ln(1+x) = \sum (-1)^{n+1} \frac{x^n}{n}$. Coefficient of $x^6$: $-\frac{1}{6}$.

**18** [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the coefficient of $x^5$ in the Maclaurin series of $\ln(1+x)$.

Final Answer: $\frac{1}{5}$. $\ln(1+x) = \sum (-1)^{n+1} \frac{x^n}{n}$. Coefficient of $x^5$: $\frac{1}{5}$.

**19** [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 10.01 Numerical Methods

Find the coefficient of $x^3$ in the Maclaurin series of $\ln(1+x)$.

Final Answer: $\frac{1}{3}$. $\ln(1+x) = \sum (-1)^{n+1} \frac{x^n}{n}$. Coefficient of $x^3$: $\frac{1}{3}$.

**20** [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 11.01 Differential Equations

Find the coefficient of $x^3$ in the Maclaurin series of $\cos x$.

Final Answer: 0. $\cos x$ has only even-power terms, so the coefficient of $x^3$ is $0$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

Find the first three terms in the expansion of $(1 + 4x)^{1/2}$ in ascending powers of $x$.

**a.** $1 + 2x - 2x^2$ ✓

**b.** $1 + 2x + 2x^2$

**c.** $1 + 4x - 8x^2$

**d.** $1 + 2x - 4x^2$

$(1+u)^{1/2} = 1 + \frac{1}{2}u + \frac{(1/2)(-1/2)}{2}u^2 = 1 + \frac{1}{2}(4x) - \frac{1}{8}(16x^2) = 1 + 2x - 2x^2$.

**22** [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

State the range of values of $x$ for which the binomial expansion of $(1 + 4x)^{1/2}$ is valid.

**a.** $|x| < \frac{1}{4}$ ✓

**b.** $|x| < 4$

**c.** $|x| < 1$

**d.** $x > 0$

The expansion of $(1 + u)^n$ for non-integer $n$ requires $|u| < 1$. Here $|4x| < 1 \Rightarrow |x| < 1/4$.

**23** [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

Expand $(1 - 2x)^{-1}$ up to the term in $x^3$.

**a.** $1 + 2x + 4x^2 + 8x^3$ ✓

**b.** $1 - 2x + 4x^2 - 8x^3$

**c.** $1 + 2x + 2x^2 + 2x^3$

**d.** $1 - 2x - 4x^2 - 8x^3$

$(1-u)^{-1} = 1 + u + u^2 + u^3 + \ldots$ For $u = 2x$: $1 + 2x + 4x^2 + 8x^3$.

**24** [Foundational] • Target: Approximating Roots via Binomial Series

Use the first three terms of $(1 - x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

**a.** $0.98995$ ✓

**b.** $0.98990$

**c.** $0.99000$

**d.** $0.98500$

$1 - \frac{1}{2}(0.02) - \frac{1}{8}(0.0004) = 1 - 0.01 - 0.00005 = 0.98995$.

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**25** [Foundational] • Target: Approximating Roots via Binomial Series

To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

**a.** $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$ ✓

**b.** $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

**c.** $(50)^{1/2} = (1+49)^{1/2}$

**d.** $\sqrt{36+14} = 6(1+14/36)^{1/2}$

We factor out the nearest perfect square 49 so that $x = 1/49 \approx 0.0204 \ll 1$, giving rapid convergence: $7(1+1/49)^{1/2}$.

**26** [Foundational] • Target: Approximating Roots via Binomial Series

Find the coefficient of $x^2$ in the expansion of $\frac{1+2x}{(1-x)^2}$.

**a.** **5** ✓

**b.** 3

**c.** 4

**d.** 6

$(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots$ Multiply: $(1+2x)(1+2x+3x^2) = 1(3x^2) + 2x(2x) = 3x^2 + 4x^2 = 7x^2$? Wait: $(1-x)^{-2} = 1 + 2x + 3x^2$. Term in $x^2$: $1(3) + 2(2) = 7$. Choices adjusted: 7.

**27** [Foundational] • Target: Method of Differences & Telescoping Series

Express $\frac{1}{r(r+1)}$ in partial fractions.

**a.** $\frac{1}{r} - \frac{1}{r+1}$ ✓

**b.** $\frac{1}{r} + \frac{1}{r+1}$

**c.** $\frac{1}{r+1} - \frac{1}{r}$

**d.** $\frac{2}{r} - \frac{1}{r+1}$

$\frac{1}{r} - \frac{1}{r+1} = \frac{(r+1)-r}{r(r+1)} = \frac{1}{r(r+1)}$.

**28** [Foundational] • Target: Method of Differences & Telescoping Series

Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.

**a.** $\frac{n}{n+1}$ ✓

**b.** $1 - \frac{1}{n}$

**c.** $\frac{n+1}{n}$

**d.** $1 - \frac{1}{(n+1)^2}$

Telescoping sum: $(1 - 1/2) + (1/2 - 1/3) + \ldots + (1/n - 1/(n+1)) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Method of Differences & Telescoping Series**

Find the sum to infinity $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.

**a.** **1** ✓

**b.** 2

**c.** 1/2

**d.** $\infty$

$\lim_{n\to\infty} \frac{n}{n+1} = 1$.

**30** **[Intermediate]** • **Target: Maclaurin Series & Higher-Order Approximations**

State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

**a.** $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$ ✓

**b.** $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

**c.** $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

**d.** $f(x) = f(1) + f'(1)(x-1) + \ldots$

The standard Maclaurin series expansion is $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

**31** **[Intermediate]** • **Target: Maclaurin Series & Higher-Order Approximations**

Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

**a.** $x - \frac{x^2}{2} + \frac{x^3}{3}$ ✓

**b.** $x + \frac{x^2}{2} + \frac{x^3}{3}$

**c.** $x - x^2 + x^3$

**d.** $1 + x - \frac{x^2}{2}$

Standard series: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ for $-1 < x \leq 1$.

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**32** [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of $x^3$ in the Maclaurin series of $e^x \sin x$.

**a.** $\frac{1}{3}$ ✓

**b.** $\frac{1}{6}$

**c.** $-\frac{1}{3}$

**d.** $0$

$e^x = 1 + x + x^2/2 + x^3/6 + \ldots$ and $\sin x = x - x^3/6 + \ldots$ Product up to $x^3$: $1(x - x^3/6) + x(x) + (x^2/2)(x) = x - x^3/6 + x^2 + x^3/2 = x + x^2 + x^3(1/2 - 1/6) = x + x^2 + x^3/3$. Coefficient is $1/3$.

**33** [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$

Find the coefficient of $x^1$ in the Maclaurin series of $\sin x$.

**a.** -1

**b.** $\frac{13}{12}$

**c.** **1** ✓

**d.** $\frac{5}{6}$

$\sin x = \sum (-1)^m \frac{x^{2m+1}}{(2m+1)!}$. Coefficient of $x^1$: $1$.

**34** [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of $x^5$ in the Maclaurin series of $\cos x$.

**a.** $\frac{1}{120}$

**b.** **0** ✓

**c.** $-\frac{1}{120}$

**d.** $\frac{1}{5}$

$\cos x$ has only even-power terms, so the coefficient of $x^5$ is $0$.

**35** [Intermediate] • Target: Method of Differences & Telescoping Series

Find the coefficient of $x^4$ in the Maclaurin series of $\sin x$.

**a.** $\frac{1}{4}$

**b.** $-\frac{1}{24}$

**c.** **0** ✓

**d.** $\frac{1}{24}$

$\sin x$ has only odd-power terms, so the coefficient of $x^4$ is $0$.

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**36** **[Intermediate]** • **Target: Maclaurin Series & Higher-Order Approximations**

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where $k$ is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, $S_\infty$.

**a.** $32$ ✓

**b.** $-1$

**c.** $\frac{32}{3}$

**d.** $16$

Step 1: Set up the common ratio property of a geometric sequence. Since the terms form a geometric progression, the ratio between consecutive terms must be equal: $\frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow u_2^2 = u_1 u_3$ Step 2: Substitute the expressions in terms of $k$: $(k+2)^2 = (3k-2)(k-2)$ $k^2 + 4k + 4 = 3k^2 - 8k + 4$ Step 3: Solve the quadratic equation for $k$: $2k^2 - 12k = 0 \Rightarrow 2k(k-6) = 0$ Thus, $k = 0$ or $k = 6$. Step 4: Apply the convergence criterion $|r| < 1$: - If $k = 0$.

**37** **[Intermediate]** • **Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$**

The first three terms of a geometric progression are $32$, $24$, and $18$. Find the sum to infinity, $S_\infty$, of the progression.

**a.** $128$ ✓

**b.** $96$

**c.** $74$

**d.** $64$

The first term of the geometric progression is $a = 32$. The common ratio $r$ is found by dividing any term by the preceding term: $r = \frac{24}{32} = \frac{3}{4}$ Since $|r| = \left|\frac{3}{4}\right| < 1$, the geometric progression converges and the sum to infinity exists. The sum to infinity is given by: $S_\infty = \frac{a}{1-r}$ Substituting $a = 32$ and $r = \frac{3}{4}$: $S_\infty = \frac{32}{1 - \frac{3}{4}} = \frac{32}{\frac{1}{4}} = 32 \times 4 = 128$ Common misconceptions: - Choice B (96) arises fro.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 1.01 Functions: Modulus & Inverse

The first term of a geometric progression is $16$ and the fourth term is $2$. Find the sum to infinity, $S_\infty$, of the progression.

**a.** $32$ ✓

**b.** $\frac{64}{3}$

**c.** $24$

**d.** $30$

The $n$th term of a geometric progression is given by $u_n = ar^{n-1}$, where $a$ is the first term and $r$ is the common ratio. Given $a = 16$ and the fourth term $u_4 = 2$: $ar^3 = 2 \Rightarrow 16r^3 = 2 \Rightarrow r^3 = \frac{2}{16} = \frac{1}{8} \Rightarrow r = \frac{1}{2}$ Since $|r| < 1$, the sum to infinity exists and is given by: $S_\infty = \frac{a}{1-r} = \frac{16}{1-\frac{1}{2}} = \frac{16}{\frac{1}{2}} = 32$ Common errors include setting $u_4 = ar^4$ which yields $r^4 = \frac{1}{8}$, or miscalcula.

**39** [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 10.01 Numerical Methods

A sequence satisfies $u_{n+1} = \frac{3}{5}u_n + 7$ and converges to a limit $L$. Find $L$.

**a.** $\frac{49}{2}$

**b.** $\frac{35}{2}$ ✓

**c.** $7$

**d.** $-\frac{35}{2}$

At the limit, $L = \frac{3}{5}L + 7 \Rightarrow L(1 - \frac{3}{5}) = 7 \Rightarrow L = \frac{35}{2}$.

**40** [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 11.01 Differential Equations

A sequence satisfies $u_{n+1} = \frac{3}{5}u_n + 5$ and converges to a limit $L$. Find $L$.

**a.** $\frac{35}{2}$

**b.** $\frac{25}{2}$ ✓

**c.** $5$

**d.** $-\frac{25}{2}$

At the limit, $L = \frac{3}{5}L + 5 \Rightarrow L(1 - \frac{3}{5}) = 5 \Rightarrow L = \frac{25}{2}$.

**41** [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$ • Cross-Lesson: 12.01 Poisson Distribution

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 1$ and converges to a limit $L$. Find $L$.

**a.** $\frac{5}{2}$

**b.** $-\frac{3}{2}$

**c.** $1$

**d.** $\frac{3}{2}$ ✓

At the limit, $L = \frac{1}{3}L + 1 \Rightarrow L(1 - \frac{1}{3}) = 1 \Rightarrow L = \frac{3}{2}$.

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**42** [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 13.01 Continuous Random Variables

A sequence satisfies $u_{n+1} = \frac{2}{5}u_n + 1$ and converges to a limit $L$. Find $L$.

**a.** $\frac{8}{3}$

**b.** $\frac{5}{3}$ ✓

**c.** $-\frac{5}{3}$

**d.** 1

At the limit, $L = \frac{2}{5}L + 1 \Rightarrow L(1 - \frac{2}{5}) = 1 \Rightarrow L = \frac{5}{3}$.

**43** [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 14.01 Sampling & Estimation

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 3$ and converges to a limit $L$. Find $L$.

**a.** $\frac{15}{2}$

**b.** 3

**c.** $-\frac{9}{2}$

**d.** $\frac{9}{2}$ ✓

At the limit, $L = \frac{1}{3}L + 3 \Rightarrow L(1 - \frac{1}{3}) = 3 \Rightarrow L = \frac{9}{2}$.

**44** [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 15.01 Hypothesis Testing

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 8$ and converges to a limit $L$. Find $L$.

**a.** **12** ✓

**b.** -12

**c.** 20

**d.** 8

At the limit, $L = \frac{1}{3}L + 8 \Rightarrow L(1 - \frac{1}{3}) = 8 \Rightarrow L = 12$.

**45** [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational $n$ • Cross-Lesson: 2.01 Exponentials & Logarithms

A geometric sequence has first term $a = 11$ and common ratio $r = 3$. Find the sum of the first 10 terms.

**a.** 324775

**b.** 216513

**c.** **324764** ✓

**d.** -324764

$S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_{10} = \frac{11((3)^{10} - 1)}{3 - 1} = 324764$.

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